

ASHWATH GURU

Data & AI Engineer

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PROFESSIONAL SUMMARY

Senior Data Engineer with 7+ years of experience designing, building, and leading large-scale batch and streaming data pipelines using Python, Scala, and SQL on Apache Spark and Kafka. Proven expertise in ETL/ELT development, cloud data platforms (AWS/GCP/Azure), and lakehouse architecture using Apache Iceberg and Delta Lake, powering distributed MPP query engines (Trino) that serve engineering, analytics, and data science teams at scale. Skilled across the full data pipeline SDLC, including workflow orchestration (Apache Airflow, Argo Workflows), CI/CD and deployment automation (Jenkins), data quality, data governance, lineage, and observability, delivering high-availability systems against strict uptime SLAs and a single reliable source of truth. Strong track record as a domain subject-matter expert, partnering with product managers, architects, and analytics stakeholders, and mentoring engineers within Agile/Scrum teams to drive scalable, production-grade data solutions.

CORE TECHNICAL SKILLS

Languages: Python, PySpark, Scala, SQL, PL/SQL

Big Data & Lakehouse: Apache Spark, PySpark, Databricks, Delta Lake, Apache Iceberg, Trino (Distributed MPP Query Engine), Apache Hive, Hadoop, HDFS, Data Lakehouse Architecture, Snowflake, AWS EMR, AWS Glue, AWS S3

Pipeline Orchestration & DevOps: Argo Workflows, Apache Airflow, Argo CD (DAG-based Workflow Orchestration), Kubernetes, Docker, Jenkins, CI/CD, Deployment Automation, Release Management, Infrastructure as Code

Streaming & Messaging: Apache Kafka, Kafka Connect, AWS Kinesis, AWS Firehose, Confluent Schema Registry, Spark Structured Streaming, Real-Time Data Processing, Event-Driven Architecture, Lambda Architecture, Kappa Architecture, OpenSearch

Cloud Platforms: AWS (S3, Glue, Lambda, IAM), Google Cloud Platform (GCP), Microsoft Azure (Azure Data Factory)

Data Quality & Observability: Instana, AWS CloudWatch, AWS Glue Schema Registry, Pipeline Monitoring & Alerting, Data Lineage, Data Quality Validation, SLA Management, Root Cause Analysis

Databases & Data Warehousing: PostgreSQL, NoSQL, Oracle, AWS Redshift, DynamoDB, AWS Athena, Redis, HBase, Data Modeling, Dimensional Modeling, Star Schema Design, Snowflake Schema Design

AI / GenAI: Large Language Models (LLMs), LangChain, OpenAI API, Hugging Face, FAISS, Retrieval-Augmented Generation (RAG), Agentic AI, Vector Databases

Practices & Methodologies: ETL/ELT Pipeline Design, REST API Design, Agile/Scrum, Data Governance, Technical Documentation, Cross-Functional Collaboration

PROFESSIONAL EXPERIENCE

Data & AI Engineer | IBM – ISL, OMS-SIP *May 2025 – Present*

- Leading end-to-end design, development, and testing of "Data Insights," a data platform built on a multi-terabyte Apache Iceberg data lake, transforming data access from manual, on-request intervention to a fully self-serve model for both business users and AI agents through a unified query-driven API layer.
- Designed a query abstraction service translating user and AI agent requests into optimized Trino queries over Iceberg tables on AWS S3, enabling sub-second response times on key metrics such as stagnant SKUs and node performance, down from manual turnaround previously taking up to 24 hours.
- Implemented a hot-cold data tiering strategy across the Iceberg lake with 90-day hot-data support, improving query performance for time-sensitive metrics while reducing storage cost by an estimated 30%.
- Architected the API layer for agentic access, defining request contracts, authentication, and rate-limiting, enabling AI agents to query the platform programmatically alongside human users, with human-in-the-loop approval for expensive queries.
- Built a self-serve batch reporting pipeline for on-demand CSV reports, cutting report turnaround time from up to 24 hours to 15 minutes, with unit and integration testing to maintain data accuracy for downstream analytics and operations teams.
- Developed distributed batch processing jobs in Python and PySpark to transform large-scale datasets for low-latency query access.
- Owned full SDLC for pipeline changes, orchestrating deployments via Argo Workflows and Argo CD on Kubernetes, containerized with Docker, backed by automated CI/CD through Jenkins.

- Integrated Apache Kafka for event-driven log streaming and OpenSearch for indexing and search analytics, extending the platform into near real-time monitoring with sub-second to low-second latency.
- Implemented observability and alerting using Instana, maintaining pipeline uptime against SLA, and used PostgreSQL for metadata, configuration, and lineage tracking to support data quality and governance.
- Mentored junior engineers on data platform architecture, Spark job design, and debugging production pipeline issues; active participant in System Design Reviews and Agile/Scrum ceremonies.

Data Science Engineer | ACL Digital (Alten Calsoft Labs) *Mar 2022 – May 2025*

Client: Barnes & Noble | Domain: Supply Chain

- Built the organization's first custom reporting framework from scratch, enabling business users to generate tailored reports directly from a legacy Order Management System, establishing a scalable pattern reused across future initiatives.
- Designed and developed a delta ingestion pipeline using AWS Glue, AWS S3, Athena and PostgreSQL to ingest and transform data from SFTP sources, powering batch ETL workflows for legacy systems with strict accuracy and uptime requirements.
- Built serverless APIs on AWS Lambda for on-demand report generation and scheduling, reducing manual reporting effort and improving turnaround time for business users.
- Improved supply chain data with AWS workflow orchestration, reducing manual intervention and pipeline failure rate.

Client: Surescripts LLC | Domain: Healthcare IT

- Developed data solutions for doctors, pharmacies, and insurance providers using Scala on GCP, building production-grade pipelines to streamline healthcare data workflows across EMR and EHR systems while ensuring compliance with healthcare data standards.

Gen AI Labs — Internal Generative AI Initiative

- Developed a computer vision pipeline using PyTorch, YOLOv5, and Roboflow to analyze retail rack imagery, detect inventory levels, and trigger automated restocking alerts, reducing manual inventory checks. Engineered commercial and open-source RAG chatbot solutions using OpenAI and Mistral, LangChain, Streamlit, and Hugging Face, with FAISS for vector storage, integrating Guardrails and Ollama to improve response accuracy and reliability for enterprise use cases.

Member of Technical Staff - Python | OpsVeda, Inc. (Logility, an Aptean company) *Oct 2020 - Mar 2022*

- Developed and optimized supply chain data solutions for a SaaS platform, including real-time monitoring, automated Python reporting workflows, reports allowing xlsx formula inputs for derived columns and data pipelines using PostgreSQL, Liquibase, and Jenkins. Contributed to core product engineering with bi-weekly feature releases, ensuring operational reliability.

M.Tech Intern - Data Engineering | Reliance Industries *Jun 2019 - May 2020*

- Improved ETL performance by 75% through optimization of Sqoop workflows and Scala code; improved query response times by 40% by tuning Hive and Phoenix configurations.

Software Engineering Trainee | Value Global, LLC *Apr 2017- May 2018*

- Built SQL & PL/SQL data solutions for clients in the Oil & Gas domain.

EDUCATION

Master of Engineering (M.E.) in Big Data & Data Analytics — Manipal School of Information Sciences, Manipal University

June 2018 – July 2020 | CGPA: 9.05/10

Bachelor of Engineering (B.E.) in E.C.E — Sri Venkateswara College of Engineering (SVCE), Anna University

August 2012 – May 2016

CERTIFICATIONS

- AWS Certified Cloud Practitioner (CLF-C02) — Oct 2024
- Databricks Lakehouse Fundamentals — Feb 2024
- LangChain & Vector Databases in Production — Oct 2023
- Dataiku ETL Advanced Designer — Jan 2023
- Oracle Certified Professional (PL/SQL Developer) — Jan 2017